

Giftson Raj D

Data Science Engineer

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PROFESSIONAL SUMMARY

Data Science and Automation Engineer with over 2+ years of multidisciplinary experience in CAD automation, BIW design, and intelligent data-driven systems. Demonstrated expertise in developing scalable data pipelines, implementing predictive analytics models, and automating engineering workflows using Python, ML, and NXOpen customization. Proficient in integrating IoT-based health monitoring systems, time-series forecasting for financial markets, and end-to-end encrypted data solutions for secure information exchange. Strong command over data engineering frameworks, statistical modeling, SQL optimization, and full-stack automation. Skilled at bridging mechanical design logic with advanced data science methodologies to deliver efficient, high-precision, and production-ready analytical solutions across healthcare, finance, and industrial automation domains.

CORE SKILLS

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|-------------------|------------------------------|------------------|---------------------------------|
| • Python | • Prompt engineering | • OOPS | • Data Engineering & Automation |
| • Json, xml & csv | • Data visualisation | • Fullstack | • Pipeline Design |
| • Documentation | • Data collection & handling | • SQL (Advanced) | • C#.Net Framework |

WORK EXPERIENCE

STUDENT INNOVATION; (Sep 8 to Till-date)

- **Healthcare:** Applied AI/ML for **early disease prediction, stress analytics**. Integrated **IoT and wearable sensor data** for real-time patient monitoring. Built **predictive models and health-insight dashboards** using Python and ML pipelines. **In progress**
- **Finance:** Developing ML pipelines for time-series forecasting and high-frequency market analysis. Engineered predictive models for stock movement and automated data workflows for financial intelligence and risk optimization. **In progress**
- **Data Privacy & Security:** Developing **end-to-end encrypted data systems** ensuring confidentiality and compliance. **In progress**

BIW-DESIGN ENGINEER; Vectra Automation Software Pvt Ltd, (Jun 2023 to Sep 2024)

- Fixtures Design Engineer in the **BIW-Testing Engineer** Designation, handling different projects (tools)
- Spotting and **troubleshooting logic errors** in code or design related to **prototyping** and **assembly**, I promptly notify developers with detailed descriptions of the issue and propose solutions.
- Debug and test customizations to ensure they are functioning correctly.
- Work handled in different **OEMs** and the automotive industry in engineering design for **Automation** drafting, mainly **Fixture Design** components.
- Executed while effectively collaborating with the design engineers & manufacturers to understand their needs and implement the **CAD customization toolbar** features that enhance their workflow.
- Work environment: **SolidWorks, CATIA V5, and SIEMENS NX 1919.**
- Provided training and support to end-user tips for maximizing on **custom toolbars** and **applications.**

PRODUCTS DEVELOPED

STUDENT DESIGNER; Innovation and product development center, MSME Incubation Hub, FX Engineering College

(Oct 2021 to May 2023)

- Clay Slurry is highly cohesive, so the **sieve** is designed with a reciprocating pressing roller to emulate the manual process.
- A **Cold dryer unit** with features to split the clay mass.
- **Clay Leather Hard State Drier**-Dries the wet pots to green pots during the forecast weather, by a controlled de-humidifier airflow and a mechanical fan system to ensure the air is thrown on all sides of the pots.

- **Decorative Power Generating Window**-Windows space utilized for architecting and power generating. Using a design Table for designing. Received a fund of **9.788,39 EUR** from MSME.
- **Agriculture Produces Dewatering and Drying Equipment**- Agricultural products are converted to value-added product like juice and flakes (pulp)
- **The dual-axis solar tracking system is used for drying seafood and agricultural products. It received a fund of 11,747.38 EUR** from MSME.

PATENT

- **Dual Axis Solar Tracking System**, Design no: 368330-001.
- **Decorative Power Window**, Design no: 404220-001.

MODEL PROJECTS

- **Onion Storage**
- **Clay De-Watering unit**
- **Ball Milling machine**
- **De-humidifier**
- Complex CAD geometry algorithms solutions.
- Identifying makes parts in a prototyping

CERTIFICATION & TOOLS

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| ● AUTOCAD (Drafting and Solid Modelling) | ● SOLIDWORKS (Advanced Mate & Mechanical Mate) |
| ● CATIA V5 (Sheet Modelling) | ● SIEMENS-NX (Customization) |

EDUCATION

- **Master's in Data Science** - ANNA University (Kumaraguru) | 2027
- **Bachelor's in Mechanical Engineering** - ANNA University (Francis Xavier) | 2023
- **Diploma in Mechanical Engineering** - DOTE (SCAD) | 2020